

Inverse Problems in Geophysics

Part 6: Regularization

2. MGPY+MGIN

Thomas Günther

thomas.guenther@geophysik.tu-freiberg.de

Recap

- singular value decomposition (SVD) as fundamental tool
- general solution: the pseudo-inverse (LS & MN special cases)
- small singular values amplify noise in the model
- for ill-conditioned (wide SV range) matrices: TSVD
 - truncate singular values at some point
- regularization: adding equations to make solution unique

Truncated singular value (TSVD) method

- Look at the singular value spectrum
- Choose a maximum number p or `rtol` and compute (e.g. by `pinv`)

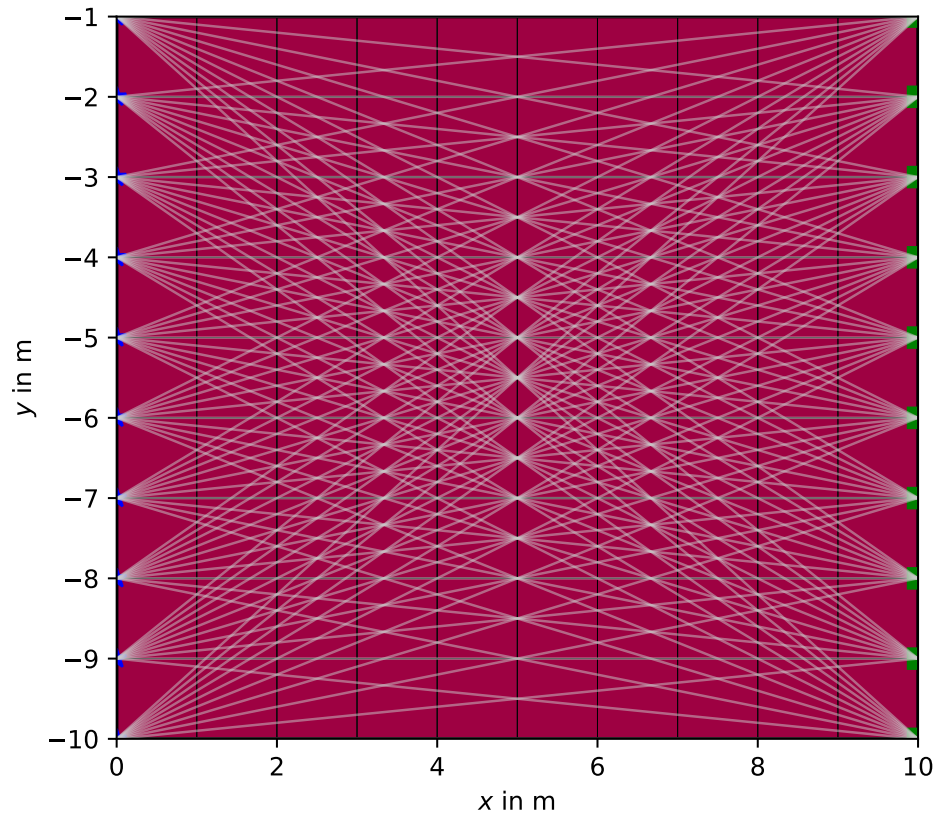
$$\mathbf{G}_p^\dagger = \mathbf{V}_p \mathbf{\Sigma}_p^{-1} \mathbf{U}_p^T$$

- How to choose p ? Trade-off between resolution and artifacts.

i Discrepancy principle (free after Occam)

Look at data fit and choose p such that the data can be fitted within the error.

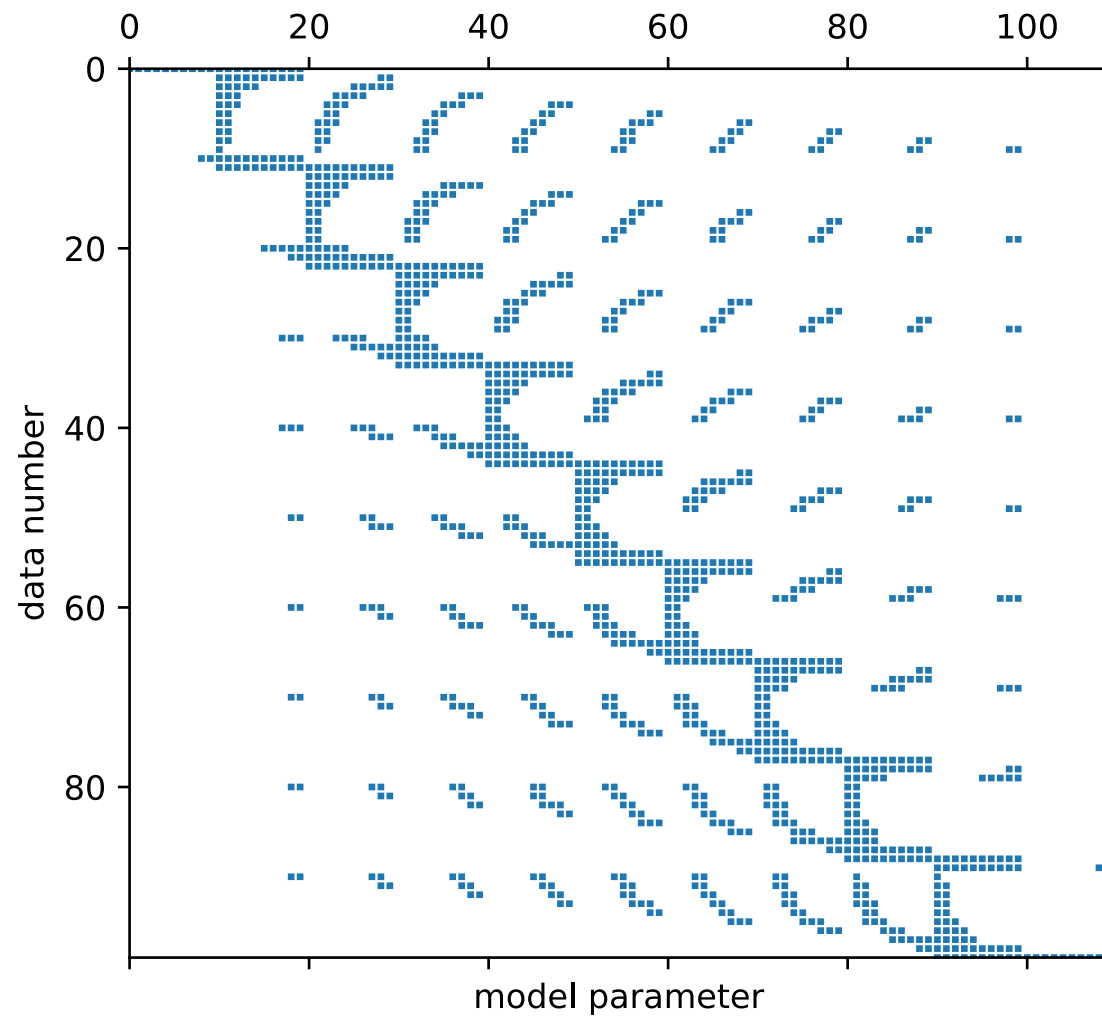
Geophysical example



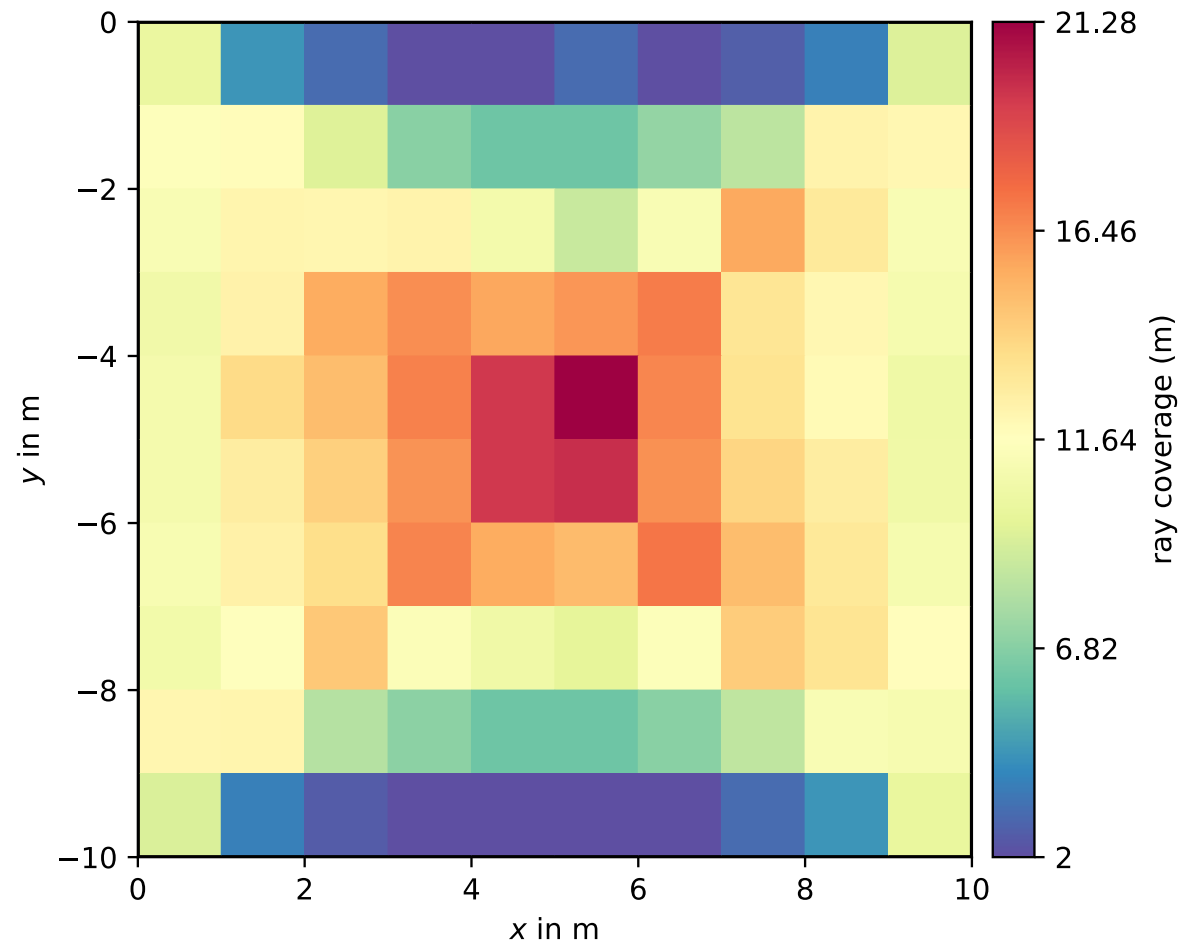
Seismic crosshole tomography (NEW)

- grid with 1m spacing (10x10 cells)
- two boreholes: shots left, geophones right, fully connected (10x10 data)
- straight ray paths (x-ray, small contrasts)

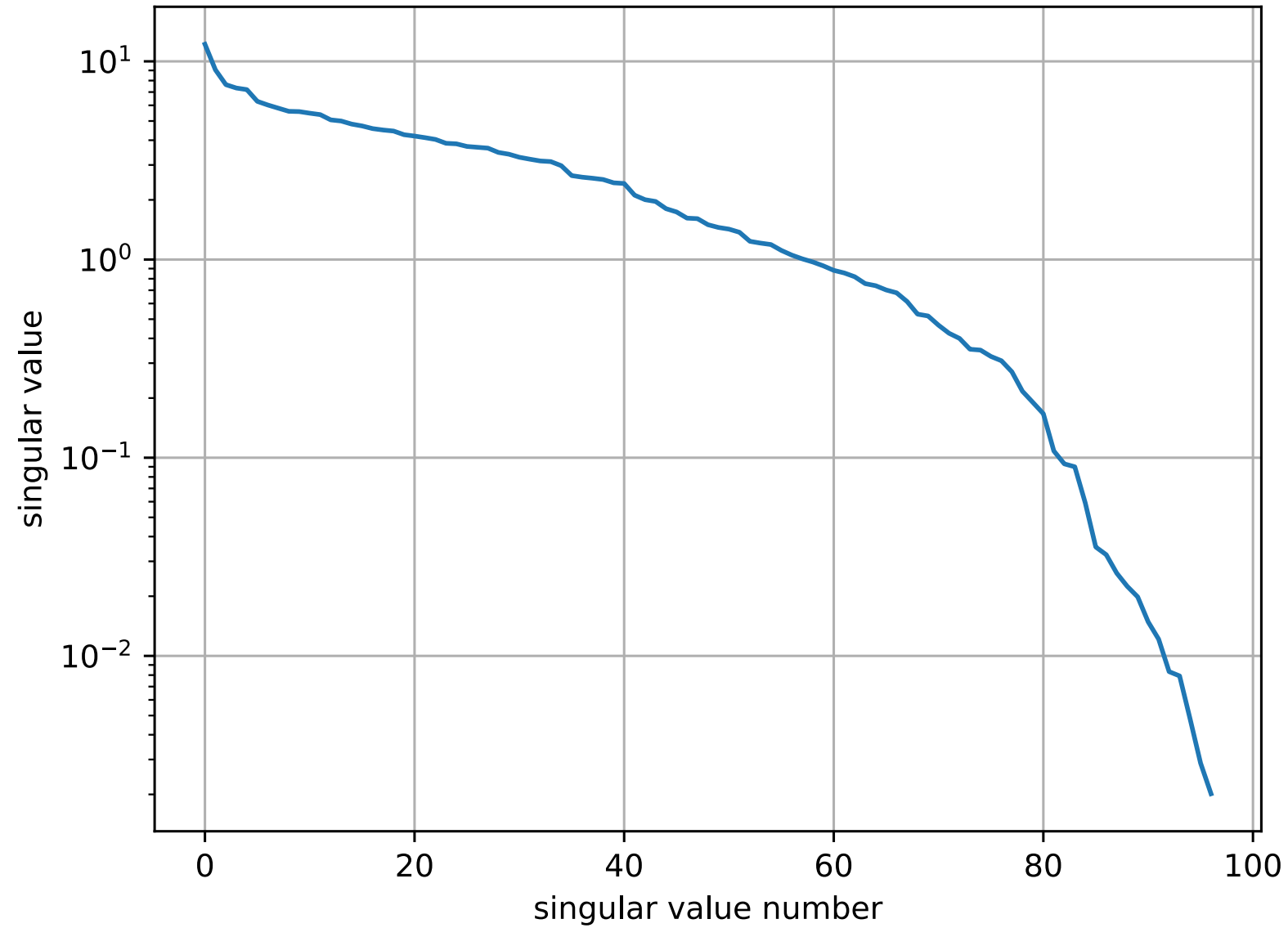
Way matrix and coverage



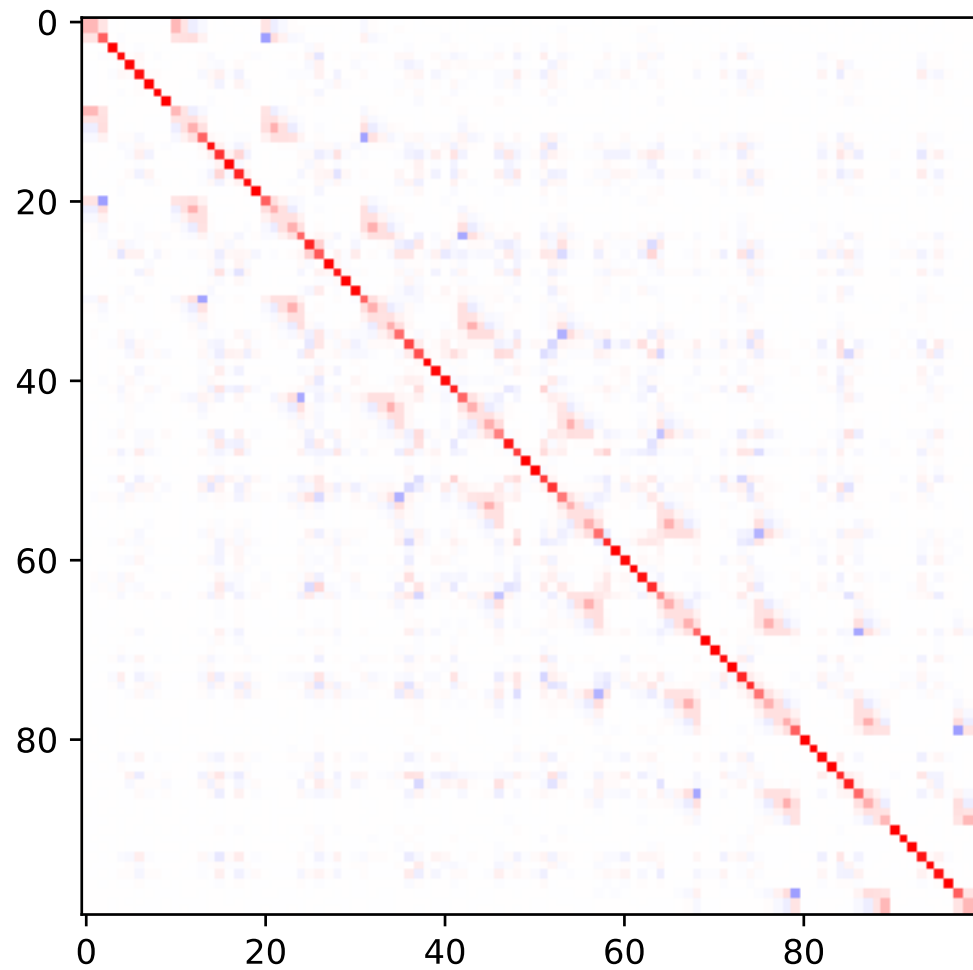
Coverage: data-sum of Jacobian



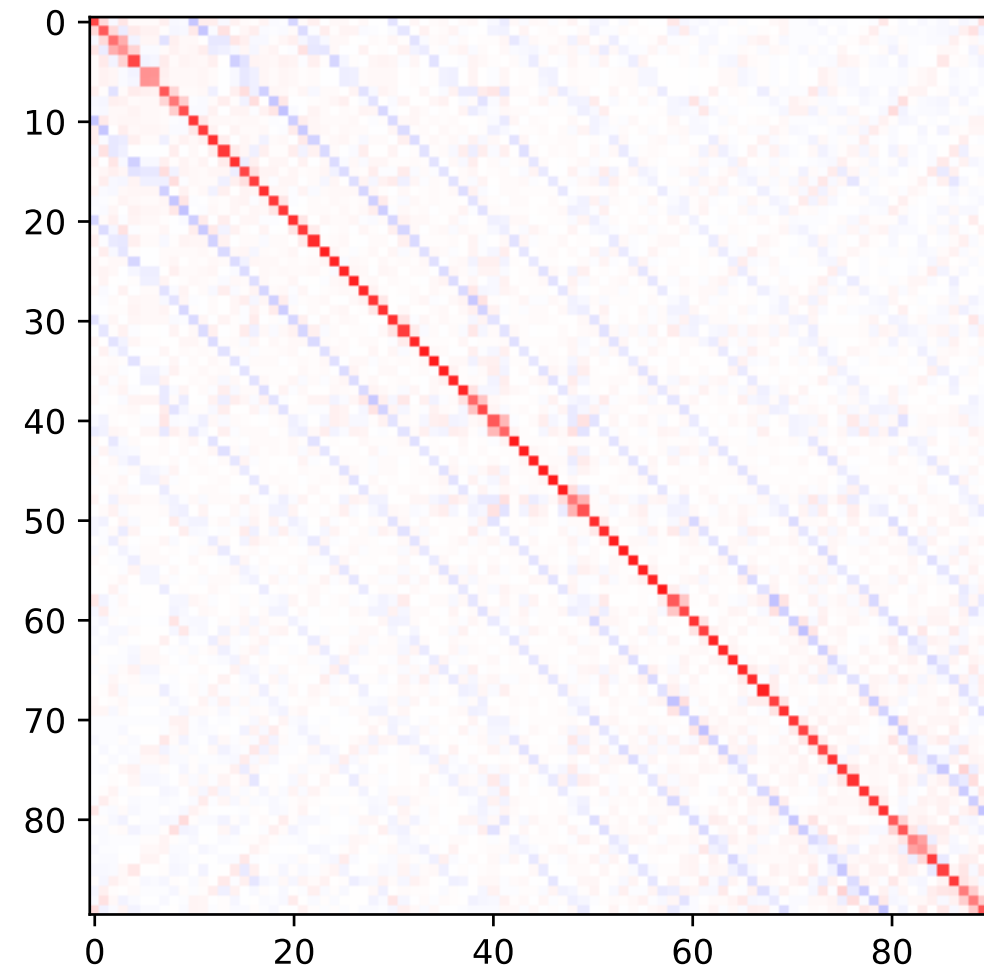
Singular value spectrum



Resolution matrices

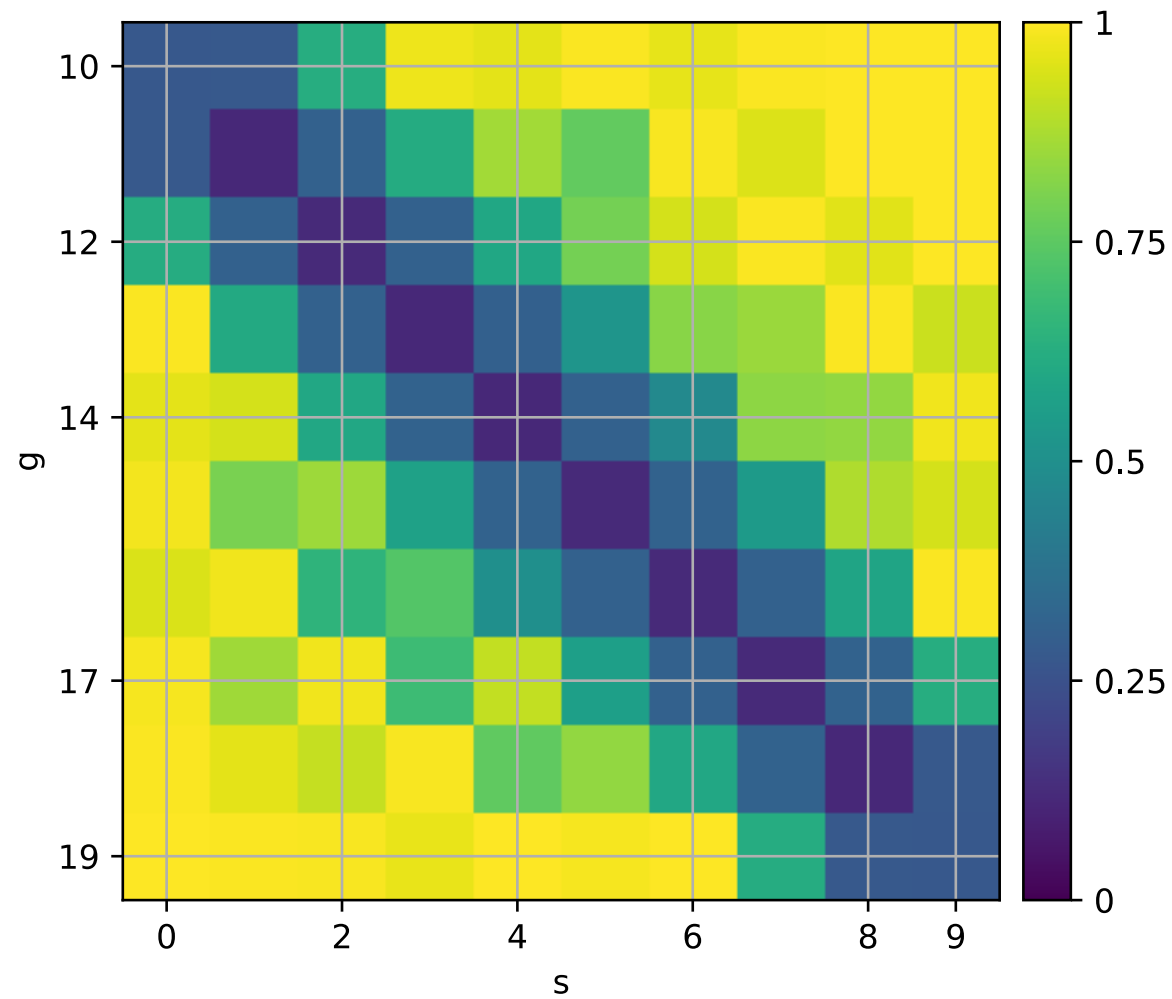


Data importance matrix

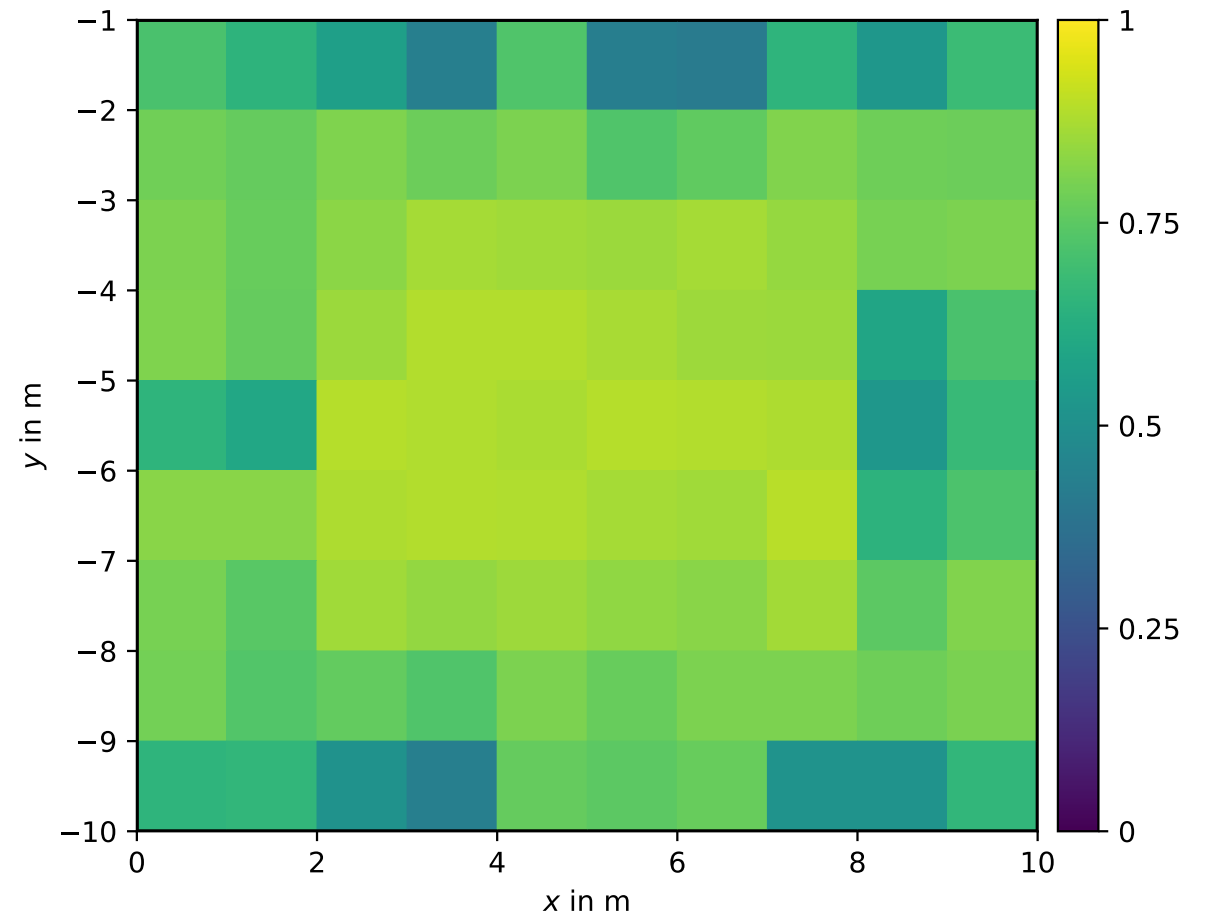


Model resolution matrix

Data importance and model resolution



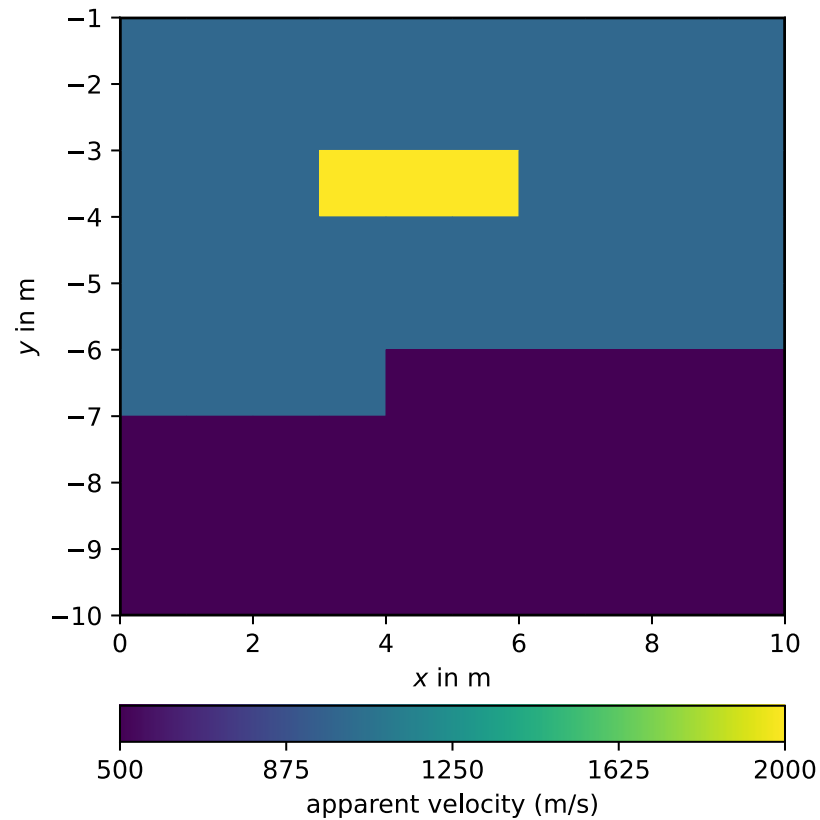
Data importance



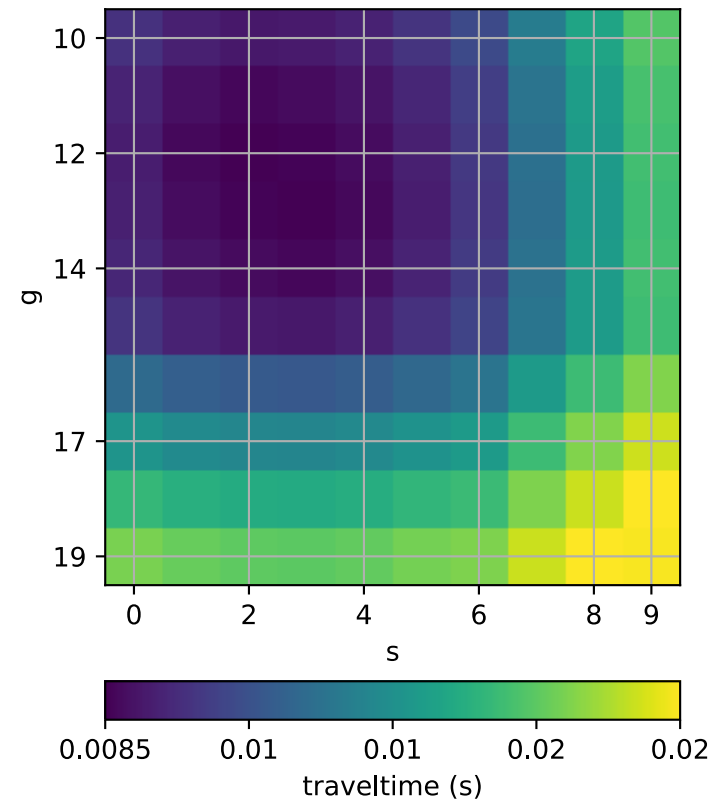
Model resolution

A synthetic model

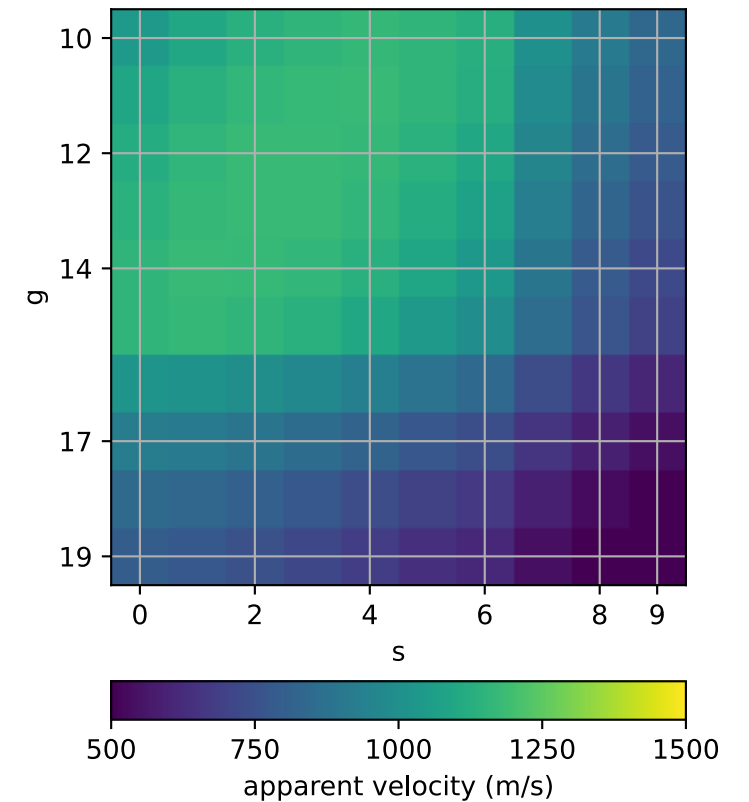
Synthetic model



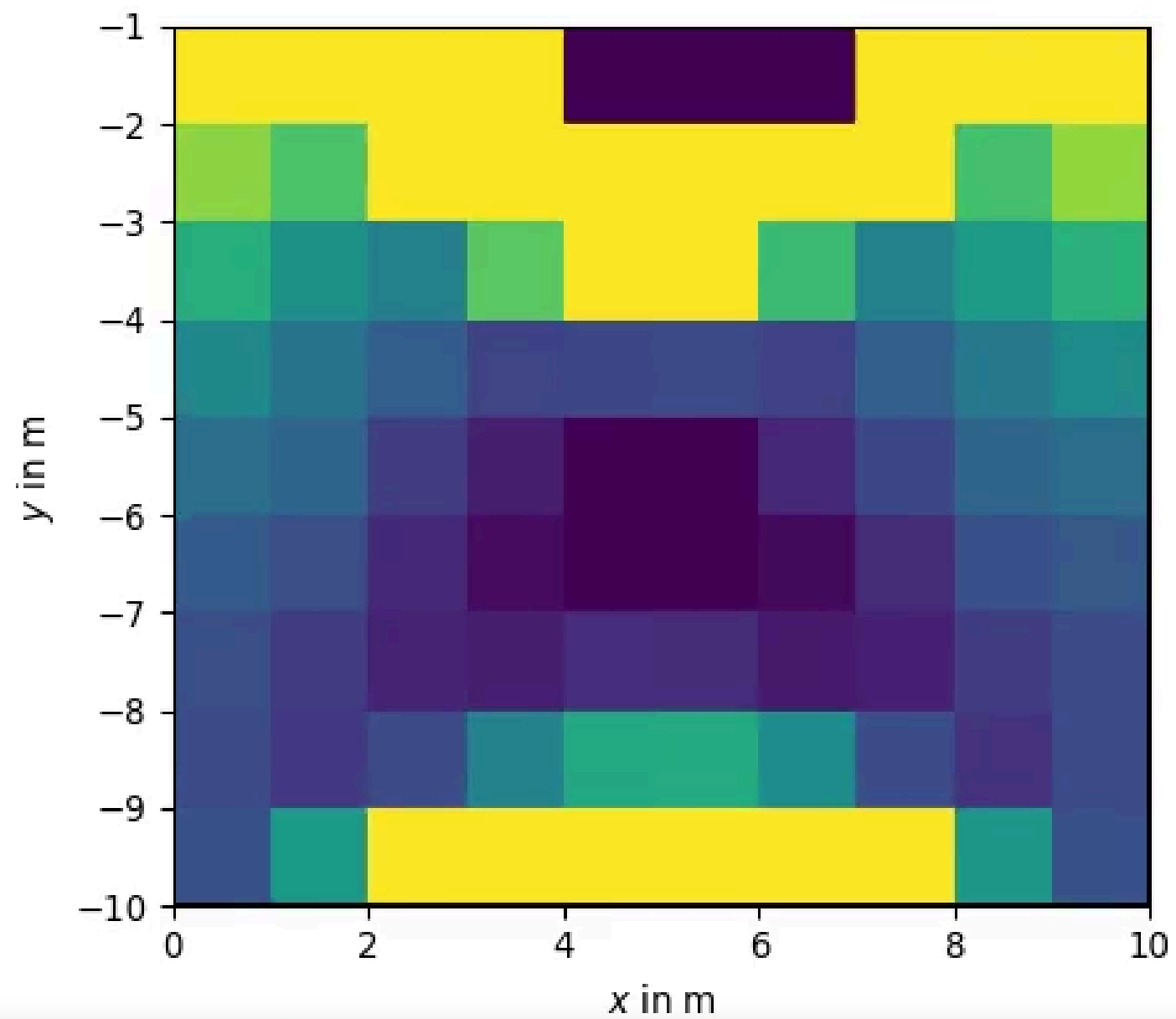
Traveltimes



Apparent velocity



Inversion with TSVD

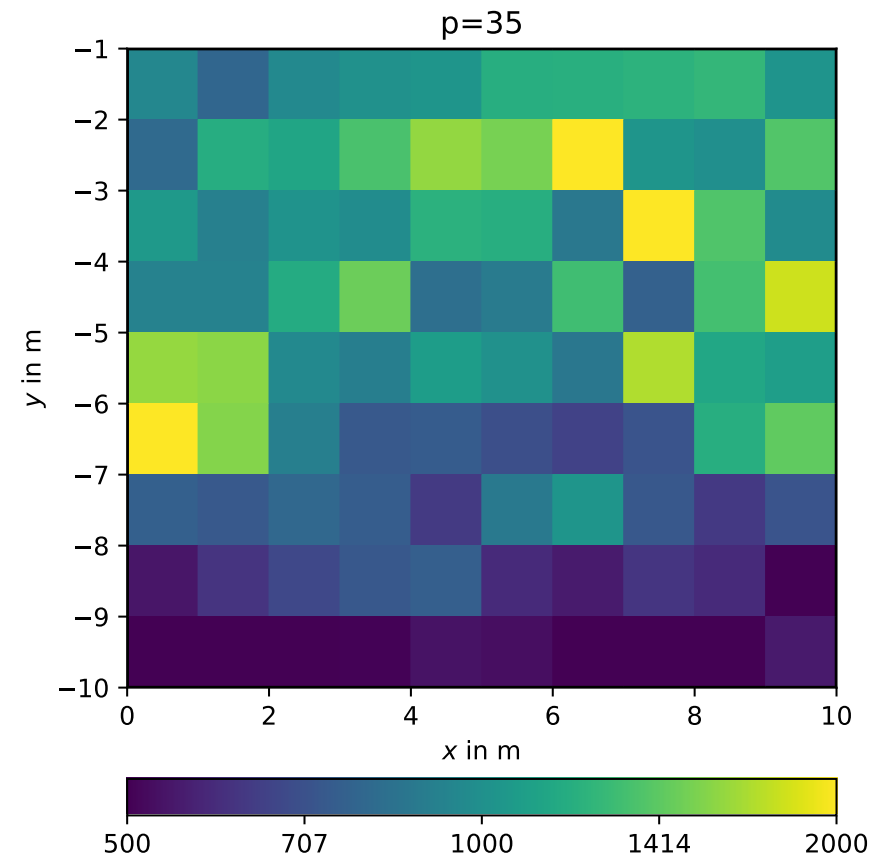


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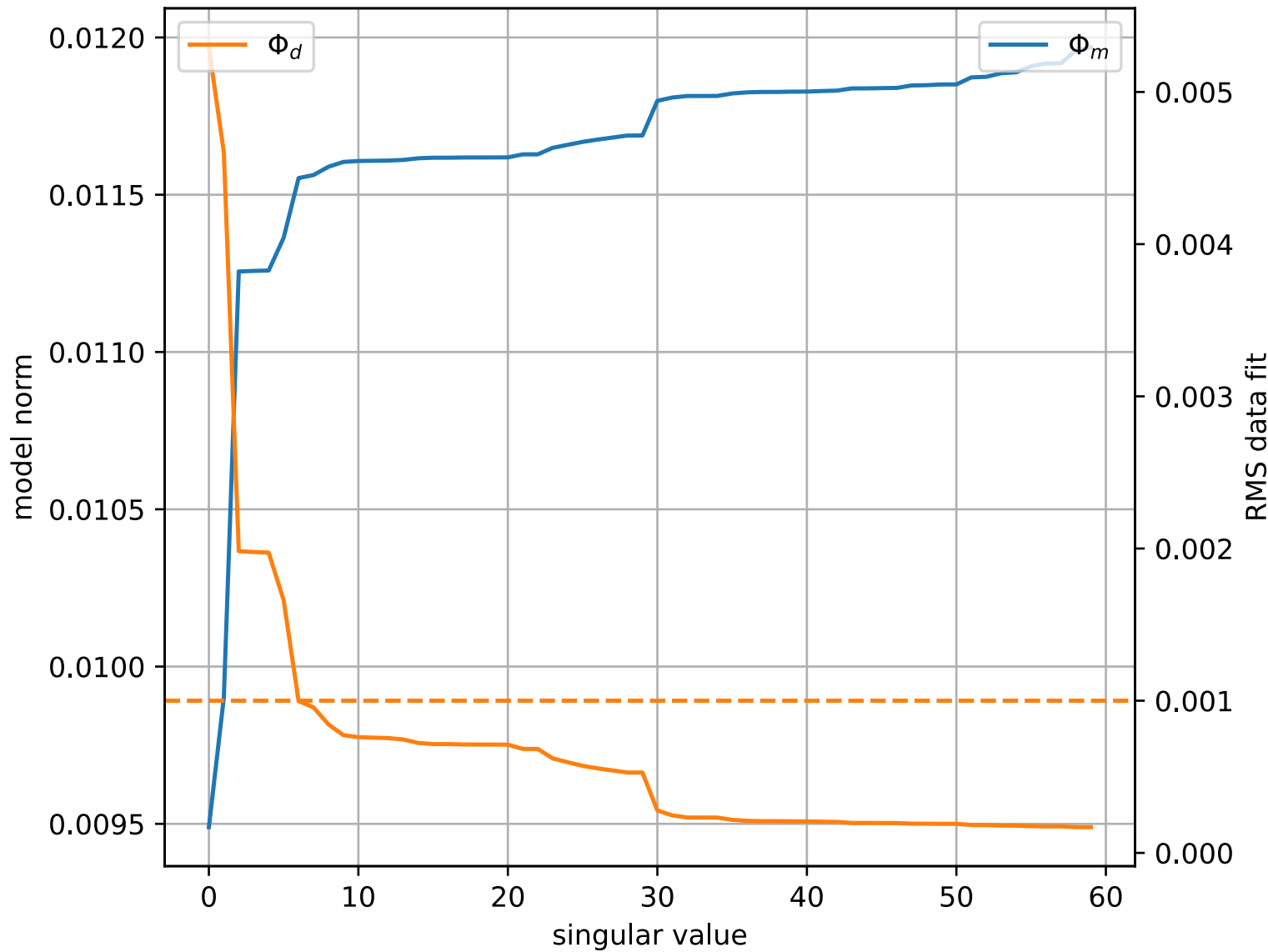
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Inversion with TSVD



Choosing p: Data and model norm



Regularization

- making under-determined and ill-posed problems unique (regular)
- make the model less sensible to small changes in the data
- adding our assumptions or knowledge (valid ranges, prior data, geostatistical behaviour)

Occams razor

Of all possible models, choose the simplest! How to define simple?

Minimum norm

All model parameters are expected to be (similarly) small

$$\tilde{\mathbf{G}}\mathbf{m} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{m} = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \end{pmatrix}$$

or close to some prior knowledge (d_3, d_4, d_5)

Smoothness constraints

Gradient (roughness) between neighboring model parameters

$$\tilde{\mathbf{G}}_{\mathbf{m}} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ 0 \\ 0 \end{pmatrix}$$

Regularization scheme

Splitting into original matrix & data and constraints

$$\tilde{\mathbf{G}} = \begin{bmatrix} \mathbf{G} \\ \mathbf{C} \end{bmatrix} \quad \text{and} \quad \tilde{\mathbf{d}} = \begin{bmatrix} \mathbf{d} \\ \mathbf{c} \end{bmatrix}$$

Damping (minimum norm)

$$\mathbf{C} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Smoothness constraints

$$\mathbf{C} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

Solving the regularized problem

Splitting into original matrix & data and constraints

$$\tilde{\mathbf{G}} = \begin{bmatrix} \mathbf{G} \\ \mathbf{C} \end{bmatrix} \quad \text{and} \quad \tilde{\mathbf{d}} = \begin{bmatrix} \mathbf{d} \\ \mathbf{c} \end{bmatrix}$$

now over-determined $\Rightarrow$ (constrained) least-squares solution

$$\tilde{\mathbf{G}}^T \tilde{\mathbf{G}} = \mathbf{G}^T \mathbf{G} + \mathbf{C}^T \mathbf{C} \quad \& \quad \tilde{\mathbf{G}}^T \tilde{\mathbf{d}} = \mathbf{G}^T \mathbf{d} + \mathbf{C}^T \mathbf{c}$$

$$\Rightarrow \mathbf{m} = (\mathbf{G}^T \mathbf{G} + \mathbf{C}^T \mathbf{C})^{-1} (\mathbf{G}^T \mathbf{d} + \mathbf{C}^T \mathbf{c})$$

Weighting data vs. constraints

$$\Phi = \|\tilde{\mathbf{G}}\mathbf{m} - \tilde{\mathbf{d}}\|_2^2 = \|\mathbf{G}\mathbf{m} - \mathbf{d}\|^2 + \|\mathbf{C}\mathbf{m} - \mathbf{c}\|^2 \Rightarrow \min$$

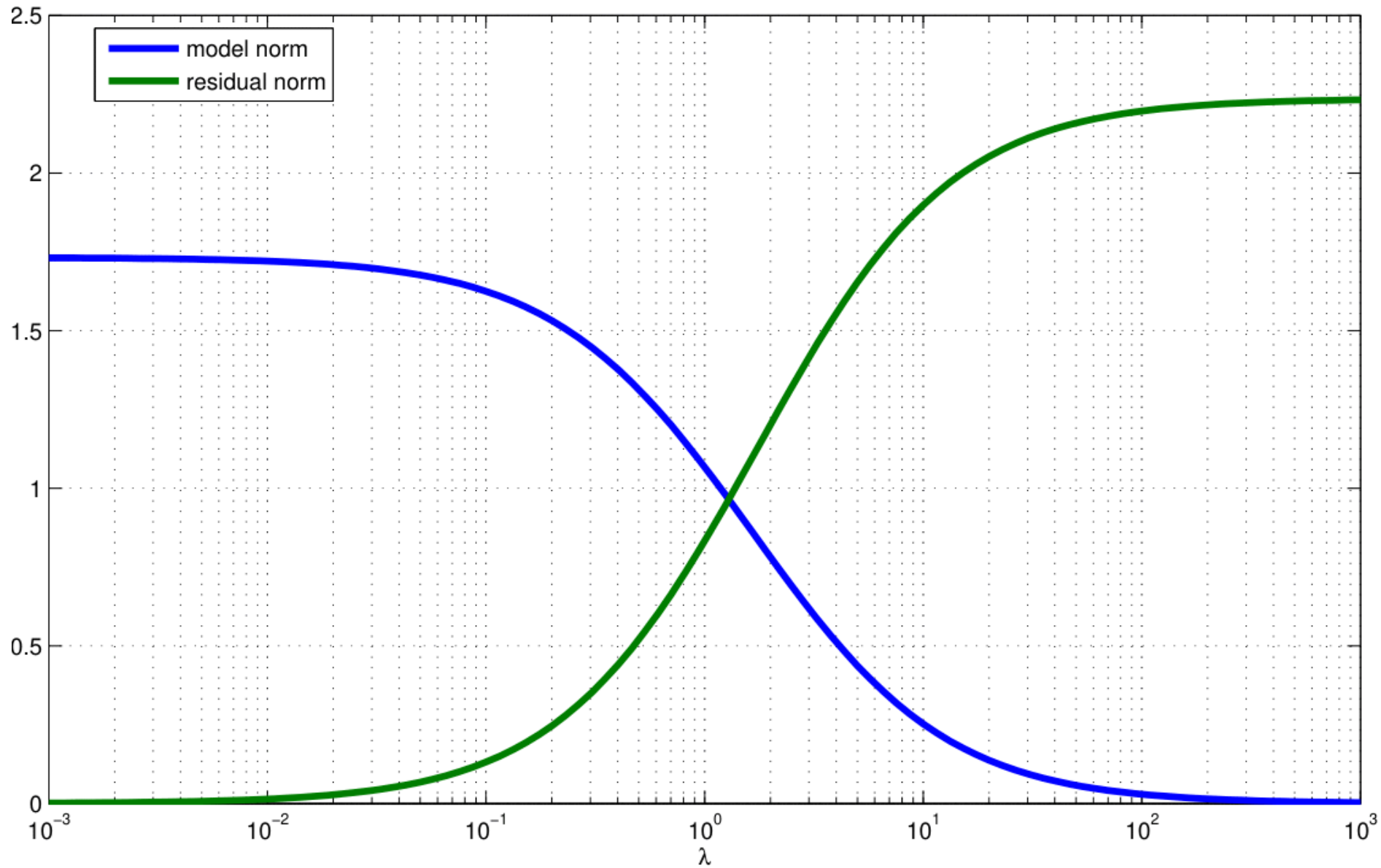
$\mathbf{d}$ and $\mathbf{c}$ have different magnitudes & units, data maybe too weak or too strong $\Rightarrow$ weighting by regularization parameter λ :

$$\Phi = \|\mathbf{G}\mathbf{m} - \mathbf{d}\|^2 + \lambda\|\mathbf{C}\mathbf{m} - \mathbf{c}\|^2 = \Phi_d + \lambda\Phi_m \rightarrow \min$$

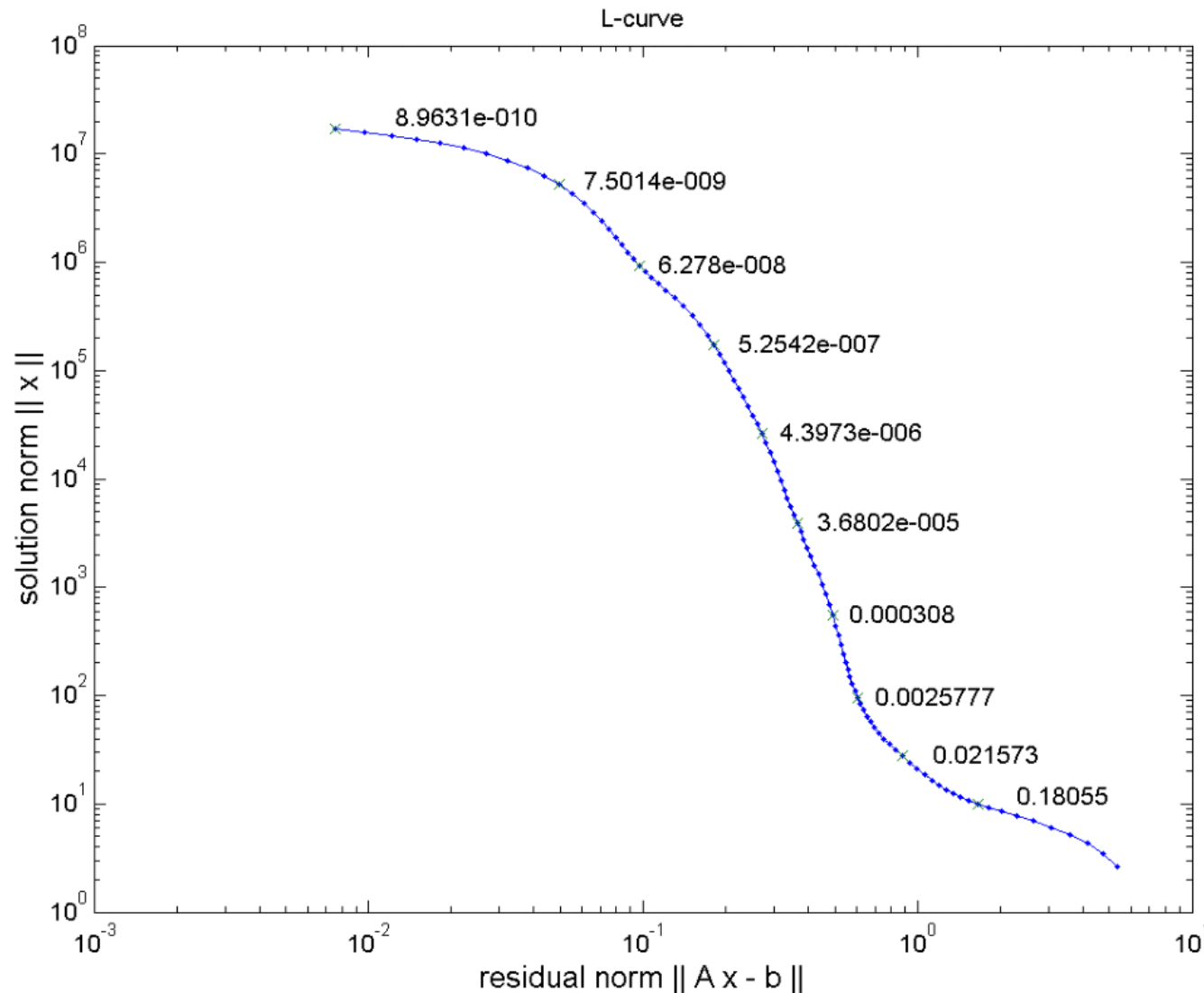
λ ..regularization strength, Φ_d/Φ_m ..data/model objective function

$$\Rightarrow \mathbf{m} = (\mathbf{G}^T \mathbf{G} + \lambda \mathbf{C}^T \mathbf{C})^{-1} (\mathbf{G}^T \mathbf{d} + \lambda \mathbf{C}^T \mathbf{c})$$

Model and data norms



The L-curve



Data vs. model norm for wide range of λ

- low data residual achieved by high norm (oscillating model)
- low model norm cannot fit the data (large misfit)
- optimum somewhere “at the corner” (not always a corner)

Choice of regularization strength

 Always have a look at your data fit and model plausibility.

- use different values and look at models (and misfit)
- try to determine the corner of the L-curve (maximum curvature)
- start large λ , decrease & stop when data misfit show no systematics

Discrepancy principle

Choose the highest λ value that is able to fit the data within errors ($\chi^2=1$)!

Know your data errors

To what extent can you explain your data?

Damped normal equations and SVD

$$\mathbf{m} = (\mathbf{G}^T \mathbf{G} + \lambda \mathbf{I})^{-1} \mathbf{G}^T \mathbf{d}$$

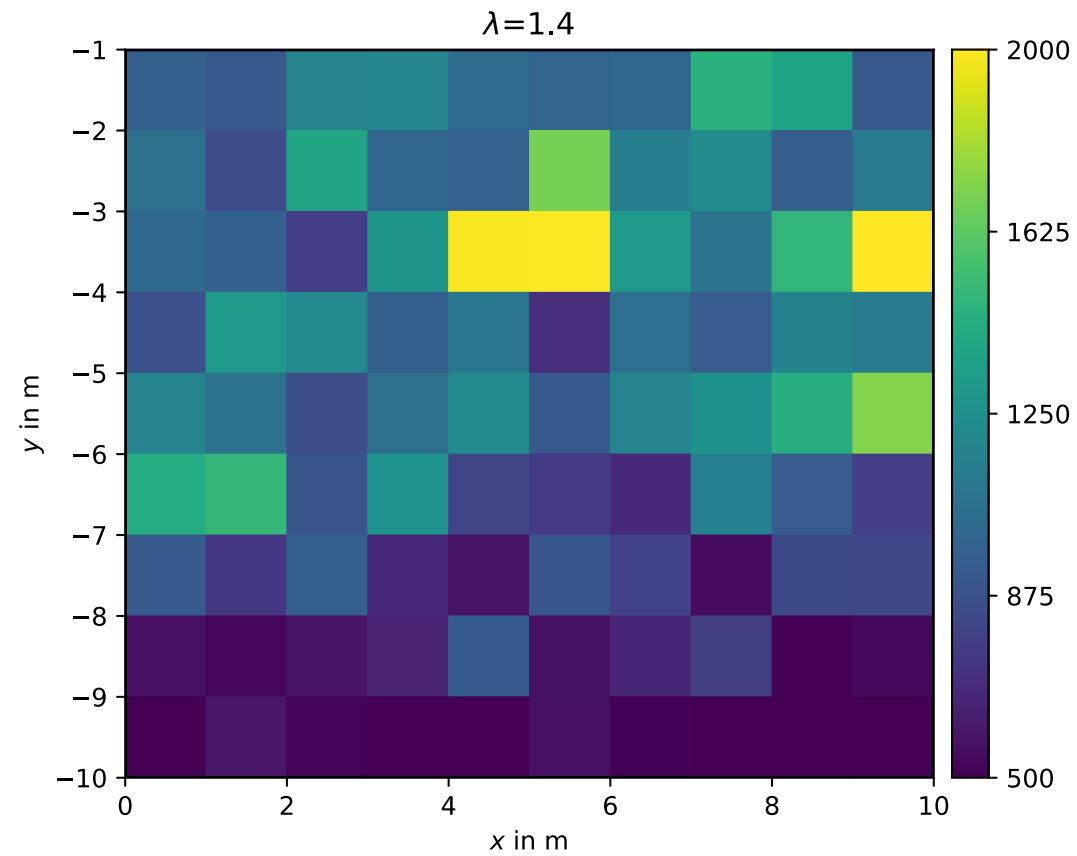
$$\mathbf{m} = (\mathbf{V} \mathbf{\Sigma} \mathbf{U}^T \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T + \lambda \mathbf{I})^{-1} \mathbf{V} \mathbf{\Sigma} \mathbf{U}^T \mathbf{d}$$

$$\mathbf{m} = (\mathbf{V} \text{diag}(s_i^2 + \lambda) \mathbf{V}^T + \lambda \mathbf{I})^{-1} \mathbf{V} \mathbf{\Sigma} \mathbf{U}^T \mathbf{d}$$

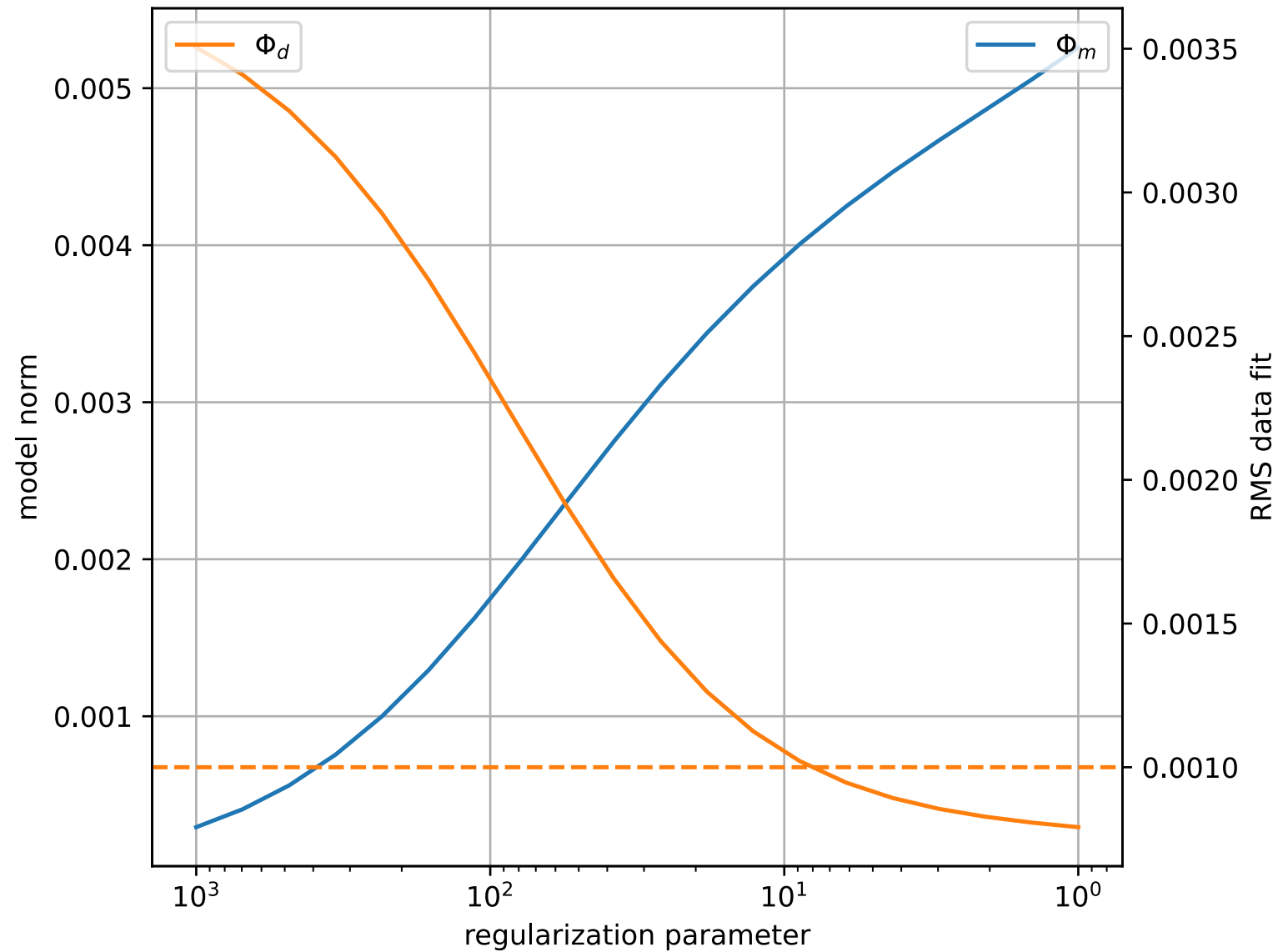
$$\mathbf{m} = \sum_i^r \frac{s_i}{s_i^2 + \lambda} \mathbf{u}_i^T \mathbf{d} \cdot v_i^T = \sum_i^r \frac{s_i^2}{s_i^2 + \lambda} \frac{\mathbf{u}_i^T \mathbf{d}}{s_i} v_i^T$$

Small singular values are damped in inversion, large unchanged

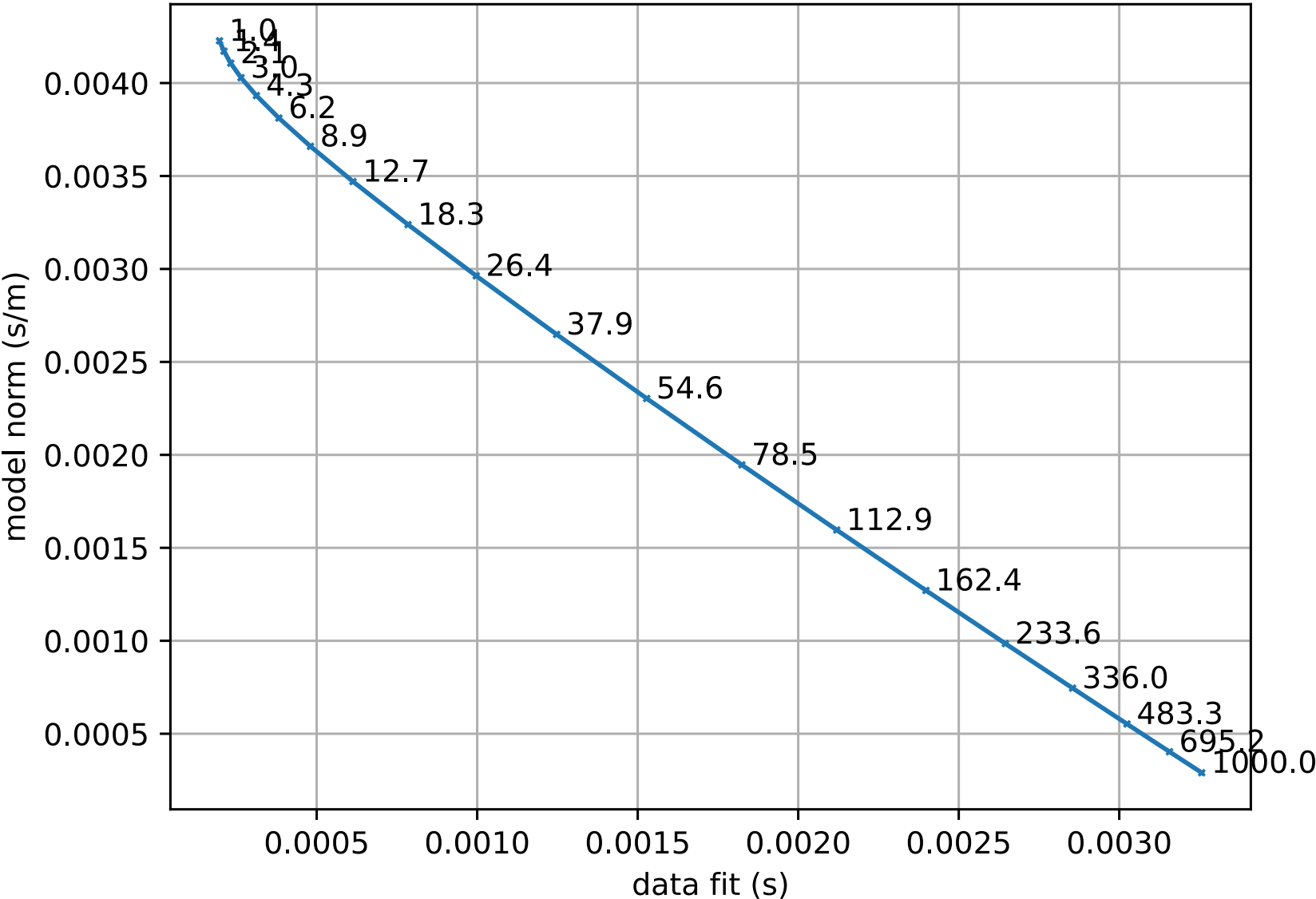
Inversion with damping



Choosing λ : Data and model norm



The L-curve



Resolution of regularized inverse problems

For $c = 0$ we have $\mathbf{G}^\dagger = (\mathbf{G}^T \mathbf{G} + \lambda \mathbf{C}^T \mathbf{C})^{-1} \mathbf{G}^T$

$$\Rightarrow \mathbf{R}^M = \mathbf{G}^\dagger \mathbf{G} = (\mathbf{G}^T \mathbf{G} + \lambda \mathbf{C}^T \mathbf{C})^{-1} \mathbf{G}^T \mathbf{G}$$

approaches $\mathbf{I}$ for $\lambda \rightarrow 0$ and deviates if λ grows

Resolution for damped normal equations

$$\mathbf{R}^M = \mathbf{V} \cdot \text{diag}(f_i) \cdot \mathbf{V}^T$$

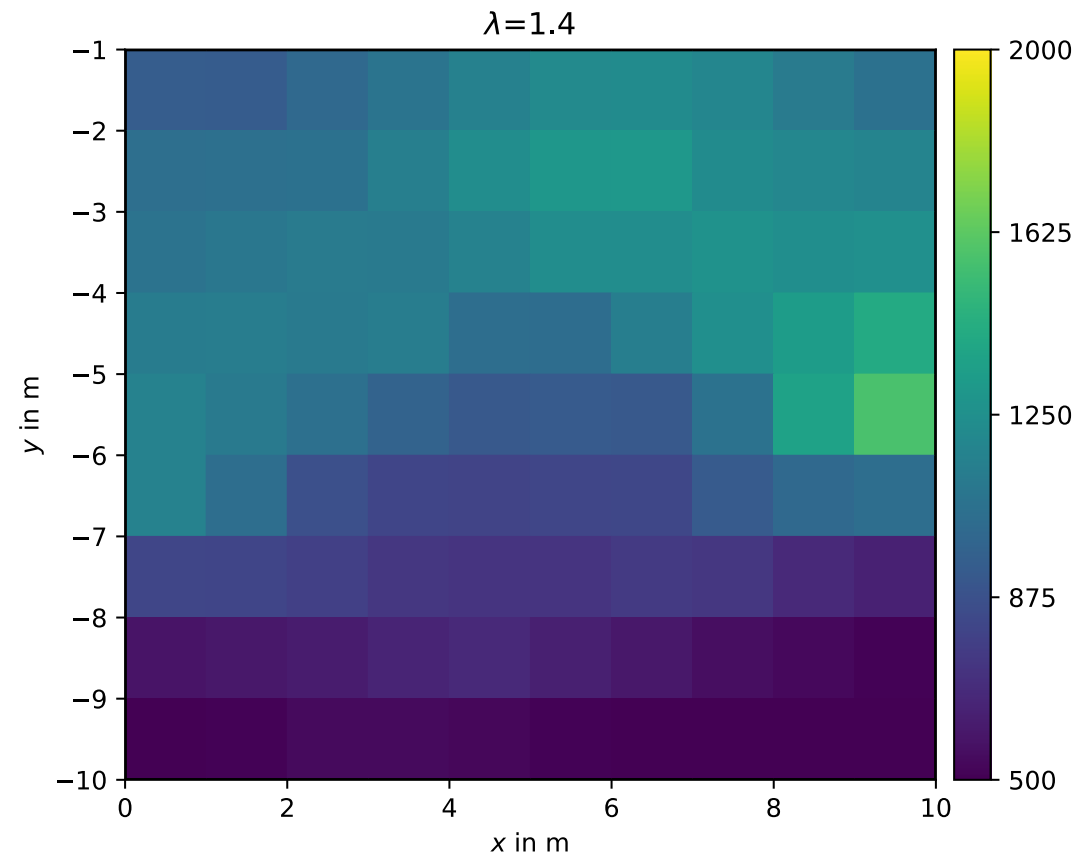
$$\mathbf{R}^D = \mathbf{U} \cdot \text{diag}(f_i) \cdot \mathbf{U}^T$$

with $f_i = \frac{s_i^2}{s_i^2 + \lambda}$ (damping of contributions from higher eigenvectors)

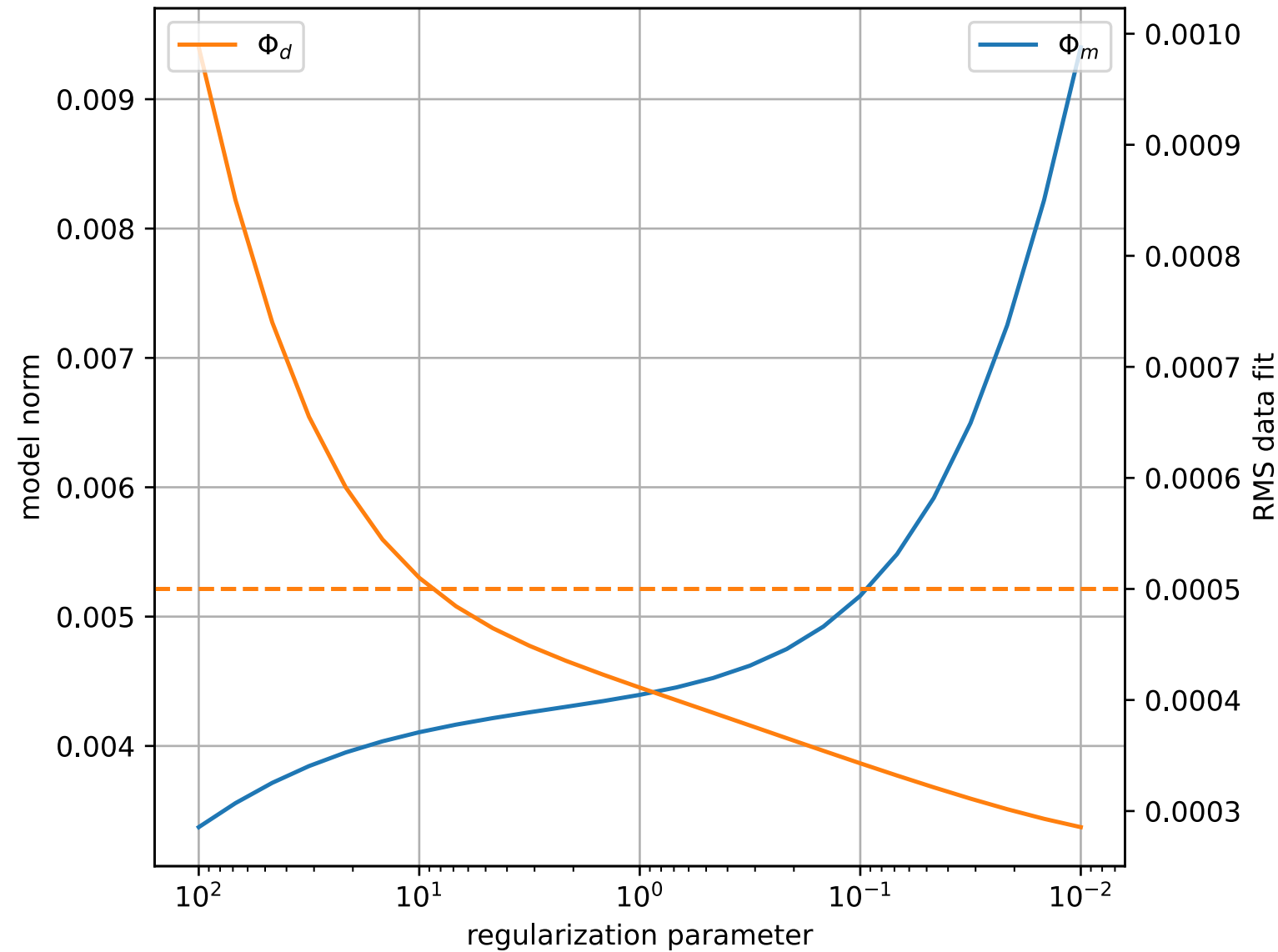
$\Rightarrow$ like for TSVD with $f_i = [1, \dots, 1, 0, \dots, 0]^T$

$$\mathbf{R}^M = \mathbf{V}_p \mathbf{V}_p^T \quad \text{and} \quad \mathbf{R}^D = \mathbf{U}_p \mathbf{U}_p^T$$

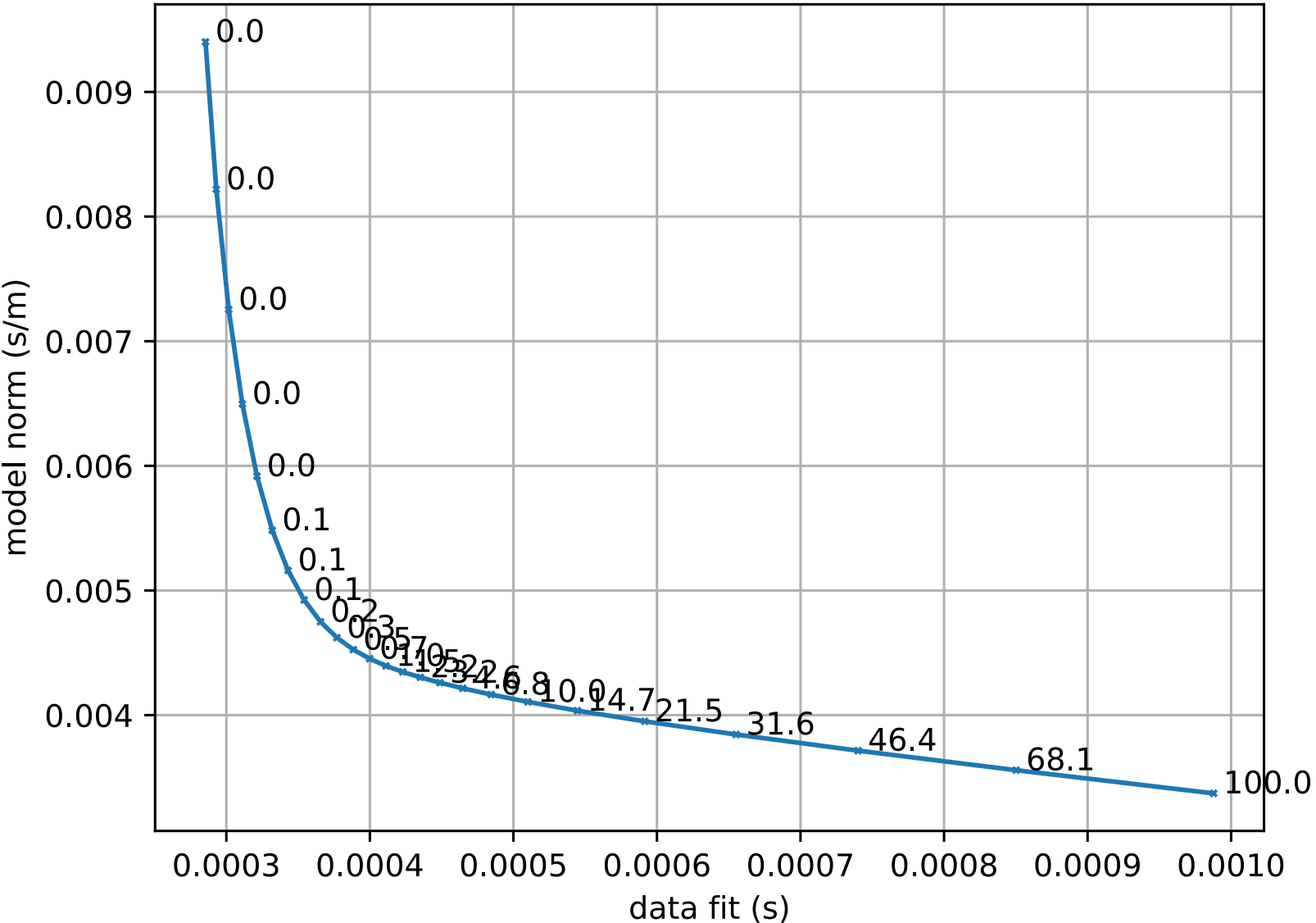
Inversion with smoothness constraints



Smoothness constraints: Data and model norm



Smoothness constraints: The L-curve

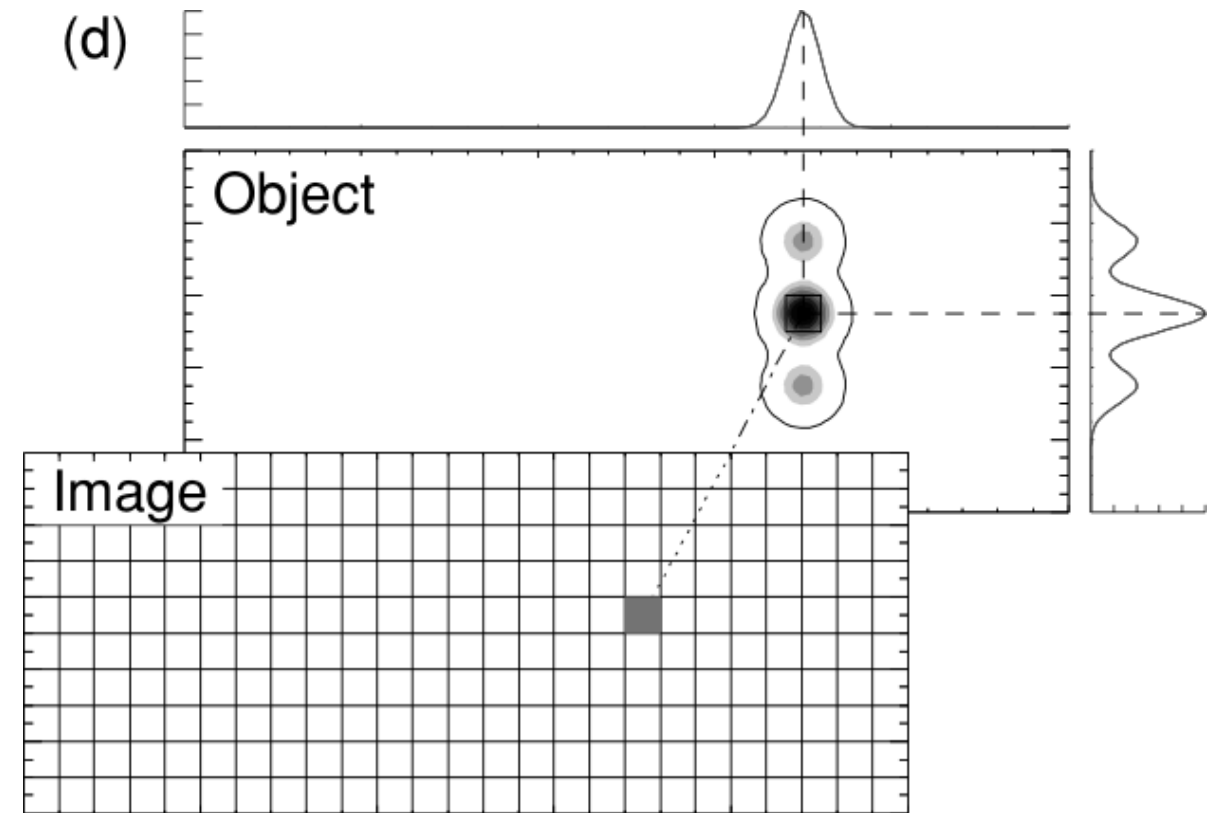


Resolution kernels (point spread functions)

$$\Rightarrow \mathbf{R}^M = \mathbf{G}^\dagger \mathbf{G} = (\mathbf{G}^T \mathbf{G} + \lambda \mathbf{C}^T \mathbf{C})^{-1} \mathbf{G}^T \mathbf{G}$$

A column represents how a cell anomaly is “spread”.

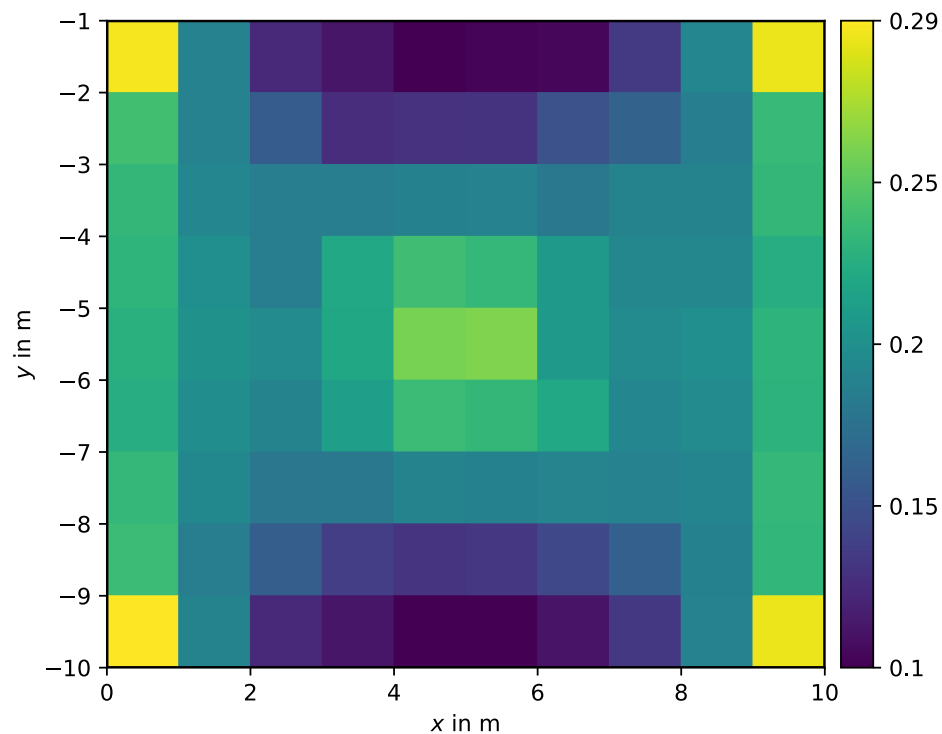
- Ideal imaging
- Contrast deficiency
- Geometrical distortion
- Side lobes



Model resolution radius

i The diagonal of the resolution matrix

states how a single model cell can be retrieved



How big needs a sphere to be to integrate a value of 1 (perfect resolution)?

Resolution radius

$$r = \frac{r_i}{\sqrt{\mathbf{R}_{ii}^M}} = \frac{\sqrt{A_i/\pi}}{\sqrt{\mathbf{R}_{ii}^M}}$$

Model covariance matrix

Let $\mathbf{x}$ be a multivariate, normal distributed random vector with mean μ and covariance $\mathbf{C}$. $\mathbf{y} = \mathbf{A}\mathbf{x}$ is also a multivariate random vector with the expectancy $E(\mathbf{y}) = \mu$ and the covariance $\text{cov}(\mathbf{y}) = \mathbf{A}\mathbf{C}\mathbf{A}^T$.

Inverse problem:

$$E(\mathbf{m}) = E(\mathbf{G}^\dagger \mathbf{d}) = \mathbf{G}^\dagger E(\mathbf{d}) = \mathbf{R}^M \mathbf{m}^{true}$$

$$\text{cov}(\mathbf{m}) = \mathbf{G}^\dagger \text{cov}(\mathbf{d}) (\mathbf{G}^\dagger)^T$$

$$\text{cov}(\mathbf{d}) = \sigma^2 \Rightarrow \text{cov}(\mathbf{m}) = \sigma^2 (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \mathbf{G} (\mathbf{G}^T \mathbf{G})^{-1} = \sigma^2 (\mathbf{G}^T \mathbf{G})$$

Modern regularization

i Regularization is key to subsurface imaging

⇒ Add any information about the subsurface:

- correlation lengths and angles
- structural boundaries (from boreholes, seismics, GPR)
- point information (samples, borehole logs,)
- limits of the possible parameters (e.g. positivity, natural bounds)

Wrap up

- SVD provides a general tool, BUT:
 - can amplify noise for ill-conditioned problems (SV spectrum)
 - truncated SVD (limiting p) can suppress this
- explicit regularization to make solution unique
 - different strategies: smoothness, minimum norm
- choice of regularization strength (λ , p) is vital